JSBL

RTGS-Integrator

Requirement Specifications 2.0

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1. Overview & Background

The transaction volume of RTGS has been significantly increased and is expected to be increased further due to the decision taken by SBP to allow customers' funds transfers for lower limits. Therefore, considering the increasing trend in RTGS transfers SBP has initiated STP project to automate the RTGS payments to minimize manual workload of customers' requests for RTGS.

This will not only help us in reducing the TAT due to duplicate efforts put in by the relevant departments due to manual posting in core banking system and PRISM but will also help us in improving our processing capabilities to manage additional transactional volume within the window allowed by SBP for RTGS payments.

Straight through Processing (STP) is an initiative used in the financial world to optimize the speed at which transactions are processed. This is performed by allowing information that has been electronically entered to be transferred from one party to another in the settlement process without the need for re-keying the same pieces of information repeatedly over the life cycle of the transaction.

About RTGS System

RTGS system started its operations in July 2008. Since its inception, enrollment of the participants is growing and volume of transactions is also increasing.

Payment System Department has also introduced Customer Transfers (MT103) and Multiple Credit Customer Transfer (MT102). This has further increased volume of transactions.

Currently, RTGS Participants are entering transactions in RTGS system manually using RTGS Client workplaces installed at their own premises. Same transactions they also have to re-enter in their Core Banking System through its own workplaces. Multiple re-keying is inefficient and error prone. Soon it will also be impractical to enter all these transactions manually due to increase in volume.

Therefore, it was decided to introduce STP (Straight Through Processing) to deal with the above mentioned issues. STP will electronically connect Participant Core Banking system with RTGS system using standard interfaces that are already provided with the RTGS system.

In this connection, Participants needs to develop an Integration Software that can pick up information from their Core Banking System and transfers it to RTGS system for processing and vice versa.

2. Bank Requirement Scope

However, since the bank has opted option of File Adapter without “Controller”; therefore it is essential to develop “INTEGRATION SOFTWARE” to cater the need of “controller” which currently provided by SBP as part of PRISM to perform multiple functions, which includes but not limited to authorization of MT102, MT103, MT202, Balance enquiry, monitoring of inflows and outflows, free format messaging and any other functionality which is currently available in the PRISM Controller provided by SBP.

The integration software should have the capability of interfacing with partner applications i.e. Remit Home System, CMS Application, T-24 and any other application which requires to be integrated with integration software.

Upon receiving the RTGS instructions from any of the partner application, the integration software, should convert the message based on its payment type (as per SBP’ prescribed formats of MT-102, MT-103 and MT-202) and push the outward MTs to PRISM by following all the relevant protocols to meet SBP’ requirements. Similarly, the integration software should have the capability to receive incoming RTGS received via PRISM from other banks for various departments/customers of the bank.

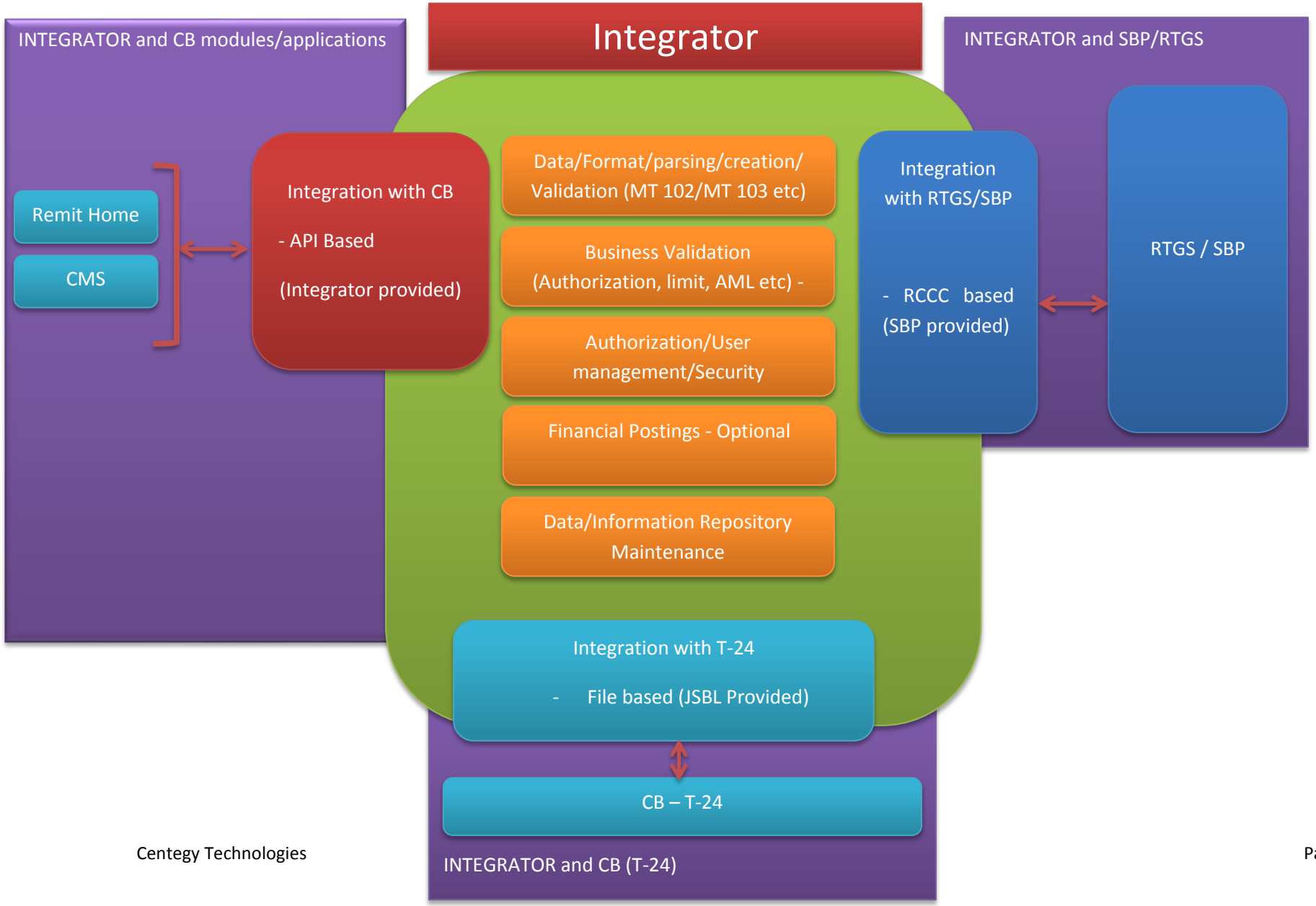
Scope Requirement by Bank

<table>
<tr><th>Ref</th><th>Module</th><th>Option-1</th><th>Option-2</th><th>Option-3</th><th>Option-4</th><th>Bank Requirement</th></tr>
<tr><td>A</td><td>Integration with Banking modules</td><td>PUSH model APIs (Integrator's own)</td><td>PULL model APIs (Client/Bank own)</td><td>File based for both (IN and OUT)</td><td>ISO messages (Phase 2)</td><td><table><tr><th>Module</th><th>Option</th></tr><tr><td>Home Remittance</td><td>1</td></tr><tr><td>CMS</td><td>1</td></tr></table></td></tr>
<tr><td>B</td><td>Integration with Financial System/T-24</td><td>APIs (Integrator's own)</td><td>APIs (Client/Bank own)</td><td>File based for both (IN and OUT)</td><td>ISO messages (Phase 2)</td><td>Option - 3</td></tr>
<tr><td>C</td><td>Integration with SBP/RTGS/PRISM</td><td>Through RCCC/API provided by SBP</td><td>File based for both (IN and Out)</td><td></td><td></td><td>Option- 1</td></tr>
</table>

SWIFT Messages compliance

Message	Required
MT-102	✓
MT-103	✓
MT-202	✓
MT-296	✓
MT-900	✓
MT-199	✓
MT-299	✓
MT-910	✓

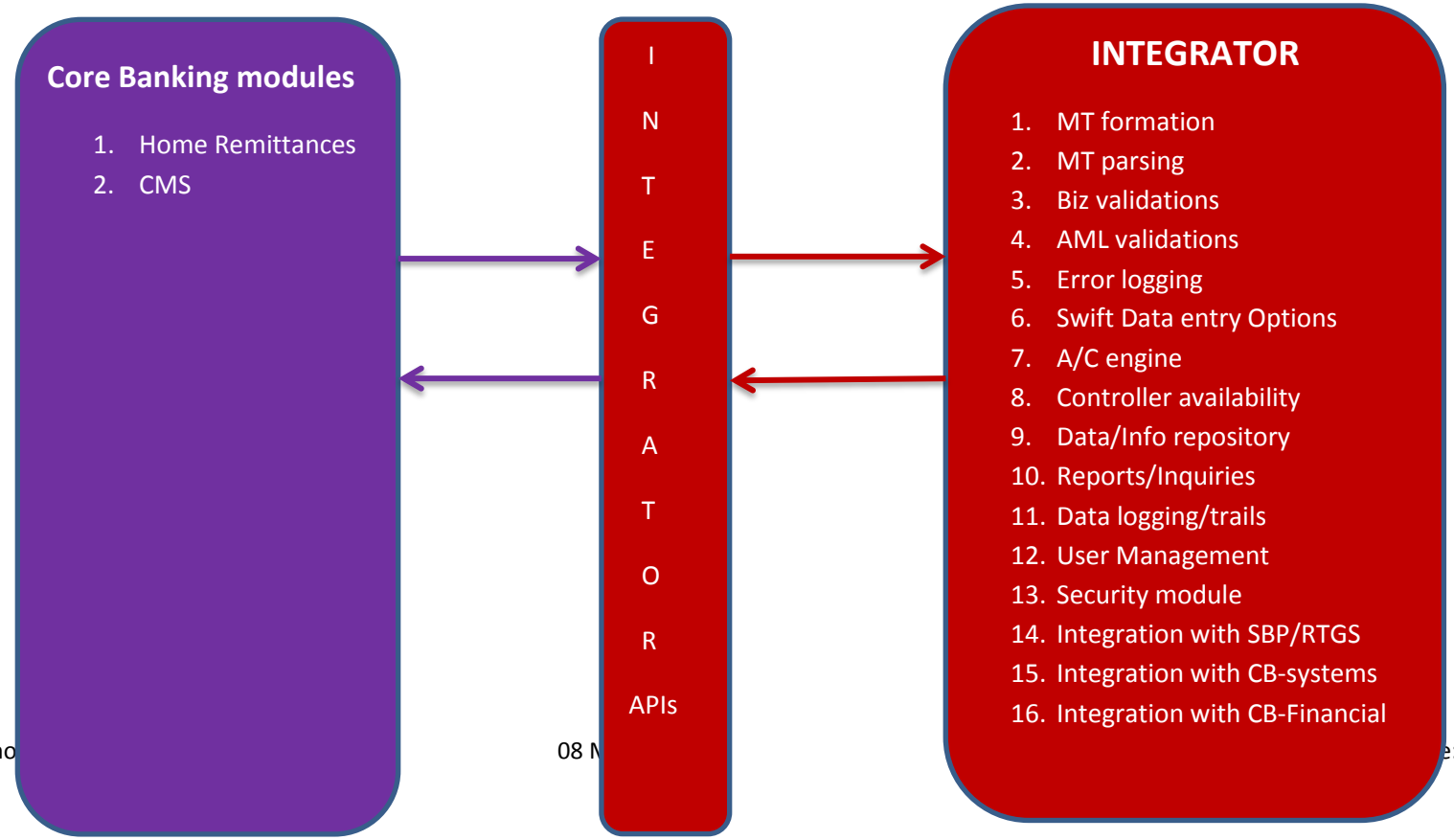
3. Architectural Diagram of INTEGRATOR



4. Integrator and Core banking modules

i- Integration through RTGS-Integrator's provided APIs

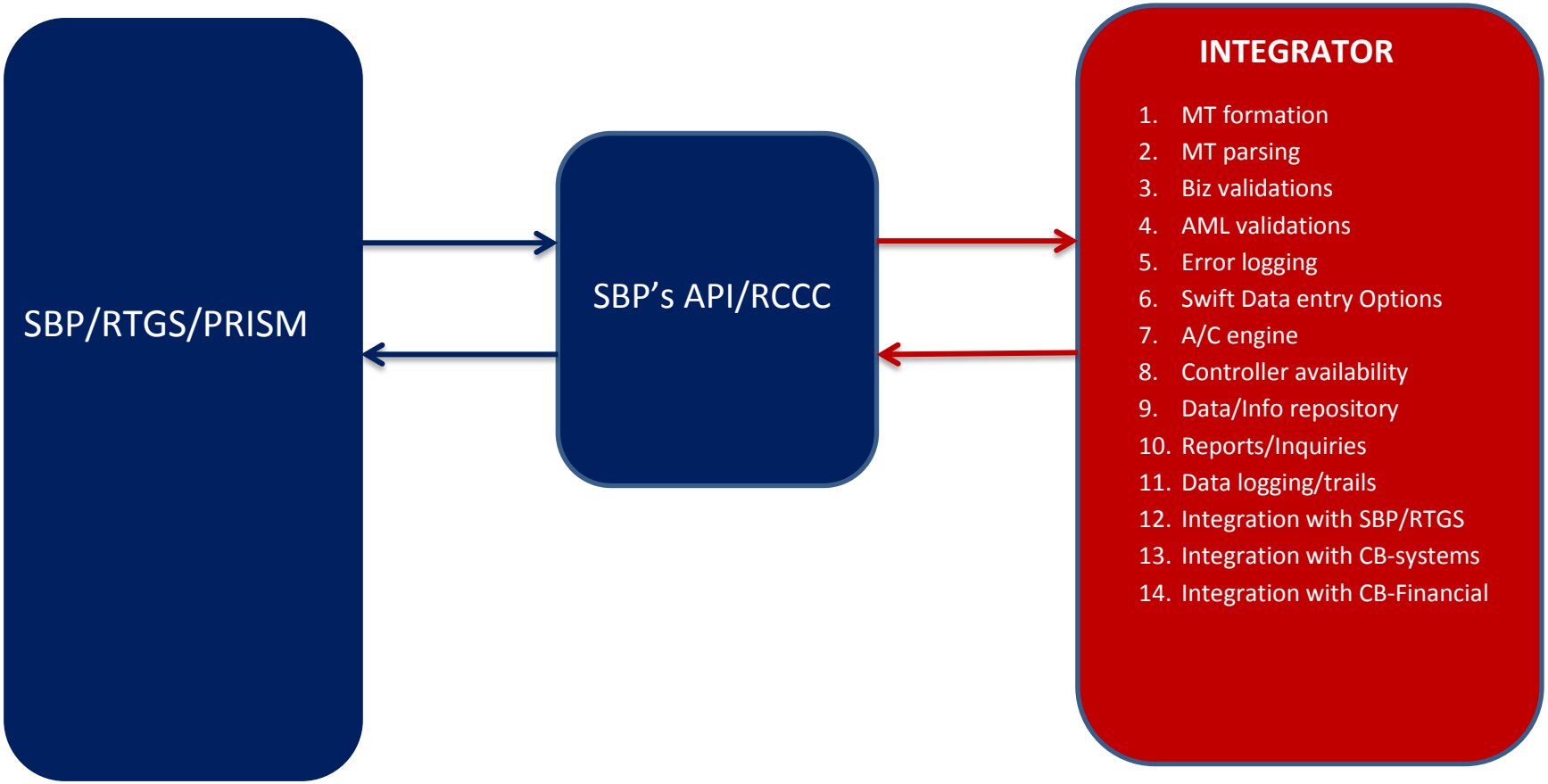
1. Core banking modules (Home remittances etc) will send data by calling API method provided by RTGS-Integrator.
2. Core banking modules will get data acceptance response from RTGS-Integrator, acknowledging that transactions have been received.
3. Core banking modules will get the status information of the data send by calling status method of RTGS-Integrator.
4. Core banking modules (Home remittances etc) will get data by calling API method provided by RTGS-Integrator.
5. Core banking modules can update the status of transactions that they have received (Incoming transactions) by calling API method.
6. **Note**: For remit home application a windows service will be developed by Centegy that will keep on checking the statuses of the data sent to RTGS-Integrator.



3. RTGS-Integrator and SBP/RTGS/PRISM

i. Integration through RTGS/PRISM/SBP provided APIs

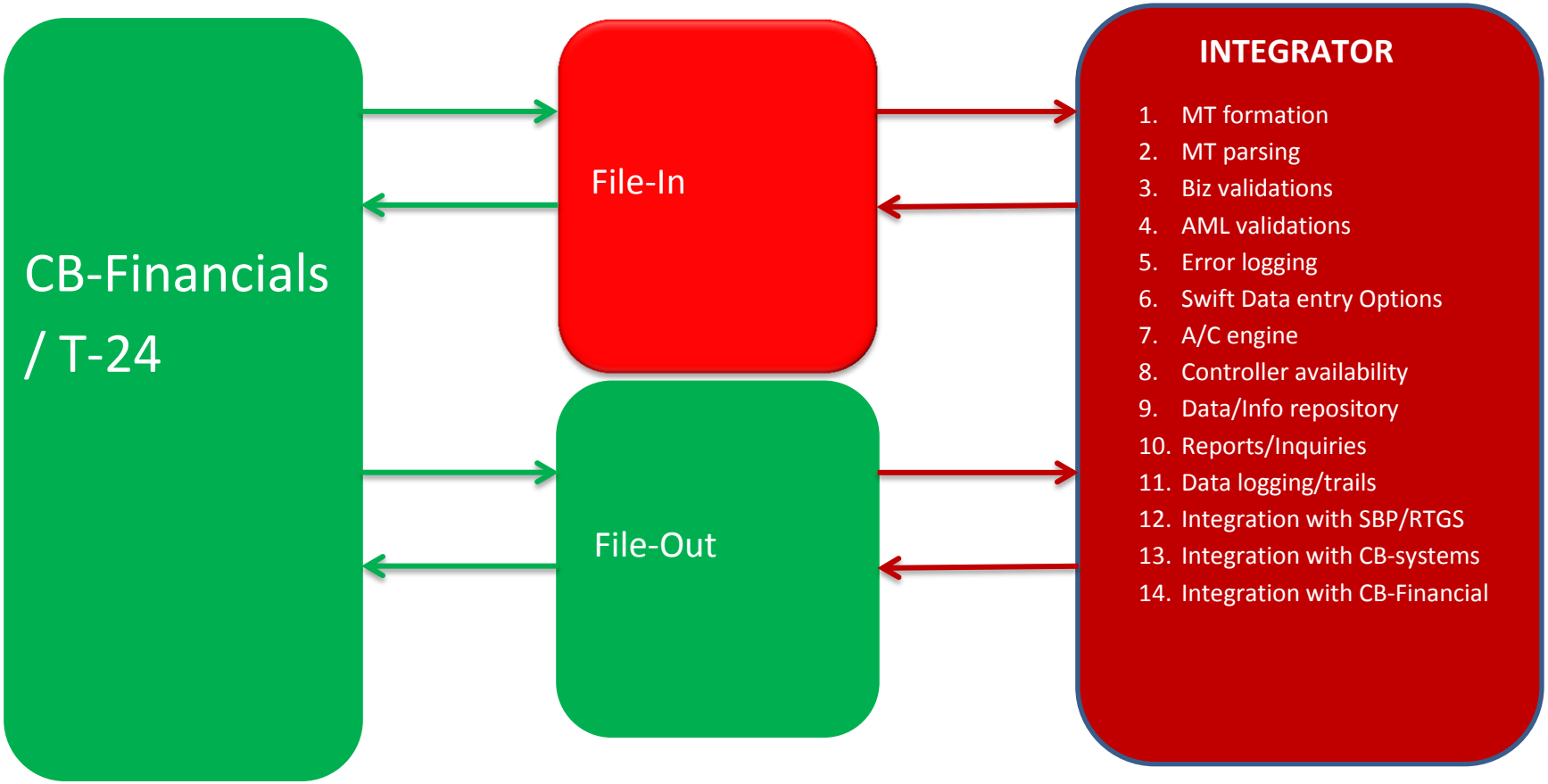
1. RTGS-Integrator will integrate with SBP/RTGS/PRISM through SBP's provided API/RCCC
2. RTGS-Integrator will send the data to SBP and update the response received from SBP.



4. RTGS-Integrator and CB Financial system

i. Integration with CB-Financial system through File Folder

1. RTGS-Integrator will generate the file according to the format specified and will place it in File folder (File IN) accessible to core banking.
2. RTGS-Integrator will access the folder (File OUT) to get the data to be sent at SBP/RTGS.



5. Specification of the RTGS Integrator

- Integrator application will expose its API methods to the banking modules like (Remit home etc).
- These banking modules will call the API methods exposed by Integrator to send and receive data.
- T-24 will be acting as a separate module/application and will integrate with Integrator through Files folders.
- Integrator will integrate with T-24 according to JSBL's provided/decided file format for sending/receiving data.
- Integrator will keep on reading File out folder to get data to be processed.
- Integrator will create a file and park it in the File In folder from where T-24 will read the data.
- There will be no acknowledgement file creation/reading concept.
- File once processed will be forwarded to the archive folder.
- Integrator will have a start of day and end of day features, before the start of day no processing will be performed and similarly after end of day activity no processing will be performed on that particular day.
- Integrator will have an optional authorization screen/layout for both Inward and outward transactions. From where user can release/amend/adjust/cancel transactions.
- Release means transactions are allowed to be processed further.
- Amend means transactions in the MT file can be amended (only demographic information).
- After amendment authorization, MT containing that particular transaction will be available at the layout mentioned at point 11.
- Adjust means forcefully ended now MT will not be processed further and will be displayed as forcefully processed.
- Cancel means no further processing.
- MT 102, 103, 202, 900, 910, 199 are the part of covered scope.
- SBP/RTGS/PRISM integration will be based on RCCC method.
- No reversal mechanism.

- If transaction is rejected/cancelled then all such transactions will not be re-processed by RTGS Integrator by any means.

6- Security and other features

No	Requirement	Centegy Response
01	The software/Application must be developed according to industry's best security practices and must comply with SBP security guidelines/regulations. The development must be done taking the Open Web Application Security Project (OWASP-2013) guidelines into account and avoid CWE/SANS TOP 25 Most Dangerous Software Errors. These guidelines are available at http://www.owasp.org	OWASP top 10 security features compliant A1 – Injection A2 – Broken Authentication and Session Management A3 – Cross-Site Scripting (XSS) A4 – Insecure Direct Object References A5 – Security Misconfiguration A6 – Sensitive Data Exposure A7 – Missing Function Level Access Control A8 – Cross-Site Request Forgery (CSRF) A9 – Using Known Vulnerable Components A10 – Unvalidated Redirects and Forwards
02	The application/software must be able to generate audit logs/trails of all events, user activities and all inbound/outbound transactions.	Refer 'RemitHome – Security Module' 1. User activity logs 2. Data changes logs
03	Service provider must agree and ensure that they will comply with the security requirements, communicated to them by JS Bank, within reasonable allocated time.	Once shared by JSB – Requirement/Observation will be Reviewed to determine scope and time efforts (as chargeable/non-chargeable)
	Technical Requirements	
04	Database	Oracle DB
05	Complete Web Based application on latest IIS version	IIS 7.0 - Dot net 4.0
06	Support of I.E, Chrome & Firefox Browsers IE	I.E Ver 11.0

07	Encrypted files wherever applicable.	File to be exchange with T-24 will be encrypted using SHA2 or Tripple DES etc.
08	Role Management Module	Refer 'RemitHome – Security Module' Menu Option's Management, assigning options to the groups
09	User Management Module	Refer 'RemitHome – Security Module' User Profile Management, assigning groups to the users.
10	Compliance of Penetration Testing Results	Once shared by JSB – Requirement/Observation will be Reviewed to determine scope and time efforts (as chargeable/non-chargeable)
11	SSL Support TLS v1.2 (bank will acquire certificate)	JSBL responsibility
12	Audit logging of each activity either manual or system generated	User information, accessing forms/options, date and time of log-in and log-out.
13	Active Cluster - In order for the high availability cluster to operate seamlessly, it's recommended that the two nodes be configured for redundancy. In other words, their individual configurations/settings must be virtually identical	JSBL responsibility

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